ASSIGNMENT REPORT-8

CLASSIFIERS WITH AND WITHOUT PCA

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For the subject

Pattern Recognition and Machine Learning

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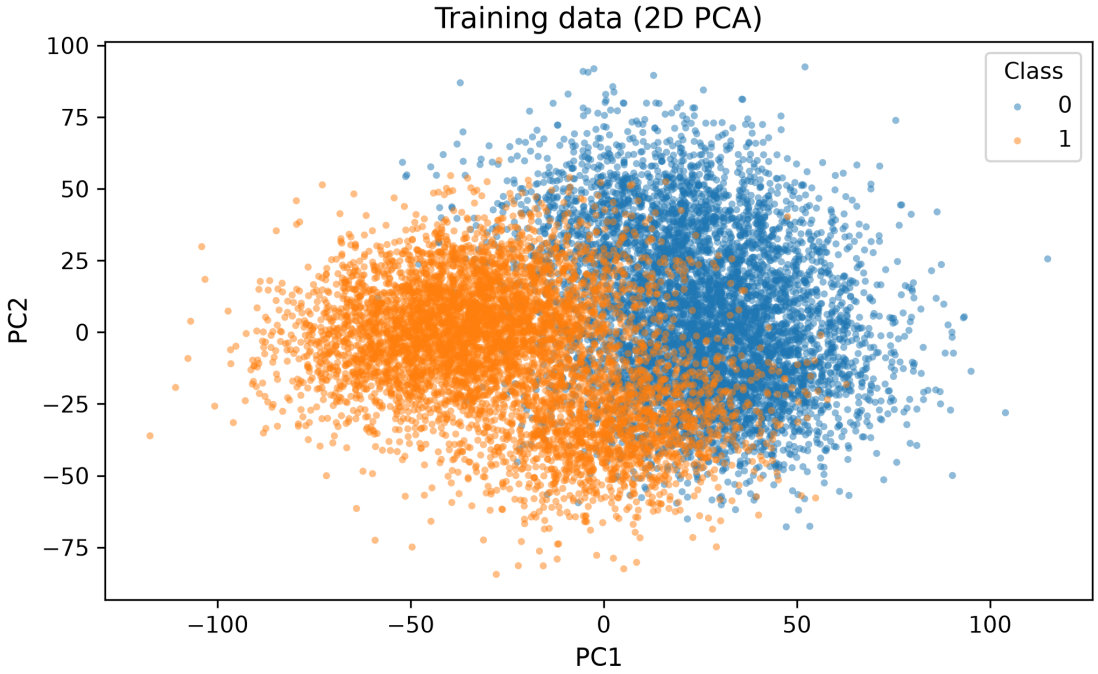
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Introduction

This assignment is all about comparing different classifiers such as GMM, KNN, LDA, SVM and also considering the reduced dimensions versions (i.e.) before and after PCA across four datasets for all 4 classifiers. For each dataset we first observe a 2D PCA version and then start building the models and record their confusion matrices and their accuracy.

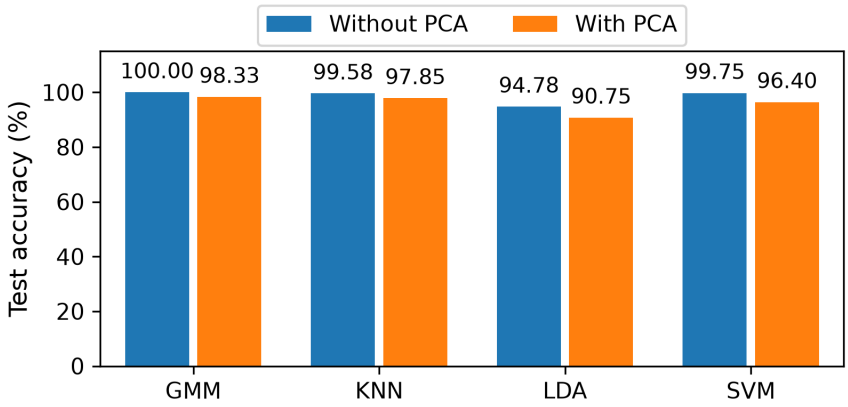
1. Synthetic data

Labels	Classes	Train	Validation	Test	Features d	PCA k
$\{0, 1\}$	2	12000	4000	4000	16	8



Accuracy (%)

	Without PCA		With PCA	
Classifier	Validation	Test	Validation	Test
GMM	100.00	**100.00**	98.40	**98.33**
KNN	99.48	99.58	97.30	97.85
LDA	94.50	94.78	91.08	90.75
SVM	99.68	99.75	96.38	96.40



Test confusion matrices Rows: actual; columns: predicted.

Without PCA

GMM	0	1
0	2001	0
1	0	1999

KNN	0	1
0	1985	16
1	1	1998

LDA	0	1
0	1880	121
1	88	1911

SVM	0	1
0	1995	6
1	4	1995

With PCA

GMM	0	1
0	1962	39
1	28	1971

KNN	0	1
0	1947	54
1	32	1967

LDA	0	1
0	1839	162
1	208	1791

SVM	0	1
0	1931	70
1	74	1925

Observations

Eventhough there is an overlap in the 2D projection, GMM still gets 100% validation and test accuracy using when we use all the 16 features. This demonstrates the capability of GMMs as it can also create non-linear boundaries as well. PCA to eight features reduces the test accuracy to 98.33%.

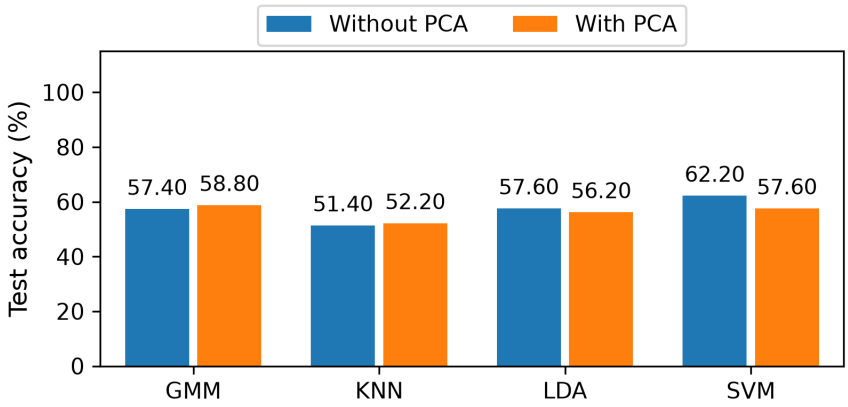
2. Image features

Labels	Classes	Train	Validation	Test	Features d	PCA k
$\{0, 1, 2, 3, 4\}$	5	2000	500	500	36	18



Accuracy (%)

	Without PCA		With PCA	
Classifier	Validation	Test	Validation	Test
GMM	62.60	57.40	63.40	**58.80**
KNN	55.60	51.40	59.00	52.20
LDA	58.80	57.60	59.60	56.20
SVM	63.00	**62.20**	64.20	57.60



Test confusion matrices Rows: actual; columns: predicted.

Without PCA

GMM

	0	1	2	3	4
0	62	7	16	9	6
1	7	75	8	5	5
2	6	3	68	15	8
3	8	4	29	37	22
4	4	3	18	30	45

KNN

	0	1	2	3	4
0	61	10	18	5	6
1	6	73	14	5	2
2	8	5	67	14	6
3	11	6	47	26	10
4	11	6	30	23	30

LDA

	0	1	2	3	4
0	66	13	12	6	3
1	7	75	9	6	3
2	8	6	64	20	2
3	13	10	31	32	14
4	7	7	16	19	51

SVM

	0	1	2	3	4
0	68	10	7	8	7
1	6	81	8	3	2
2	7	6	61	13	13
3	11	7	22	34	26
4	5	2	14	12	67

With PCA

GMM

	0	1	2	3	4
0	62	14	10	10	4
1	5	77	6	7	5
2	5	4	72	13	6
3	11	5	35	33	16
4	7	3	14	26	50

KNN

	0	1	2	3	4
0	63	12	15	7	3
1	6	80	10	2	2
2	10	5	64	13	8
3	18	5	43	24	10
4	17	7	26	20	30

LDA

	0	1	2	3	4
0	66	11	11	7	5
1	7	75	8	7	3
2	7	6	65	19	3
3	14	8	34	26	18
4	7	10	17	17	49

SVM

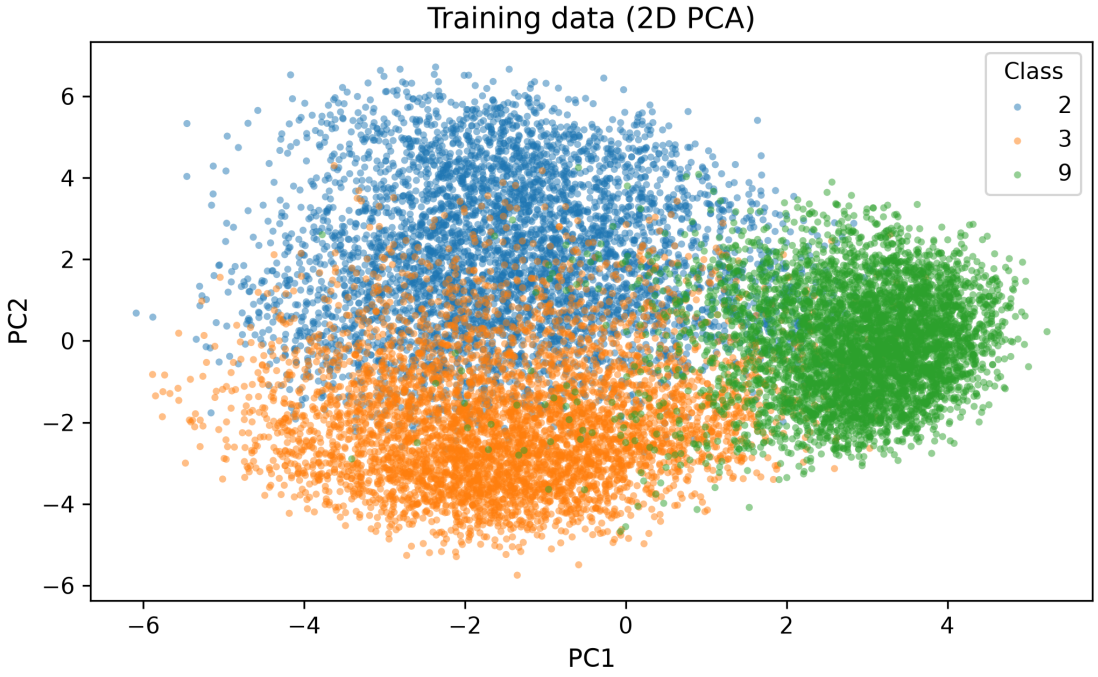
	0	1	2	3	4
0	63	13	8	11	5
1	6	76	6	9	3
2	9	6	53	18	14
3	8	8	30	31	23
4	7	3	12	13	65

Observations

Here we see PCA slightly improves GMM and KNN test accuracy, but on the other hand reduces the same for LDA and SVM. SVM performed best without PCA; GMM performed best along with PCA.

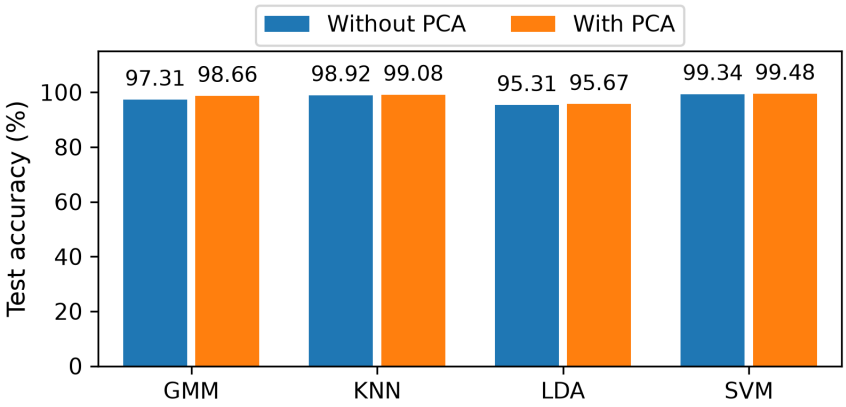
3. MNIST

Labels	Classes	Train	Validation	Test	Features d	PCA k
$\{2, 3, 9\}$	3	14424	3614	3051	784	50



Accuracy (%)

	Without PCA		With PCA	
Classifier	Validation	Test	Validation	Test
GMM	96.65	97.31	98.12	98.66
KNN	98.59	98.92	98.84	99.08
LDA	94.27	95.31	93.86	95.67
SVM	99.11	**99.34**	99.25	**99.48**



Test confusion matrices Rows: actual; columns: predicted.

Without PCA

GMM	2	3	9
2	1006	21	5
3	12	989	9
9	5	30	974

KNN	2	3	9
2	1027	2	3
3	2	998	10
9	5	11	993

LDA	2	3	9
2	987	24	21
3	32	955	23
9	12	31	966

SVM	2	3	9
2	1029	1	2
3	3	1004	3
9	3	8	998

With PCA

GMM	2	3	9
2	1022	5	5
3	4	998	8
9	5	14	990

KNN	2	3	9
2	1027	1	4
3	2	998	10
9	1	10	998

LDA	2	3	9
2	983	28	21
3	31	955	24
9	6	22	981

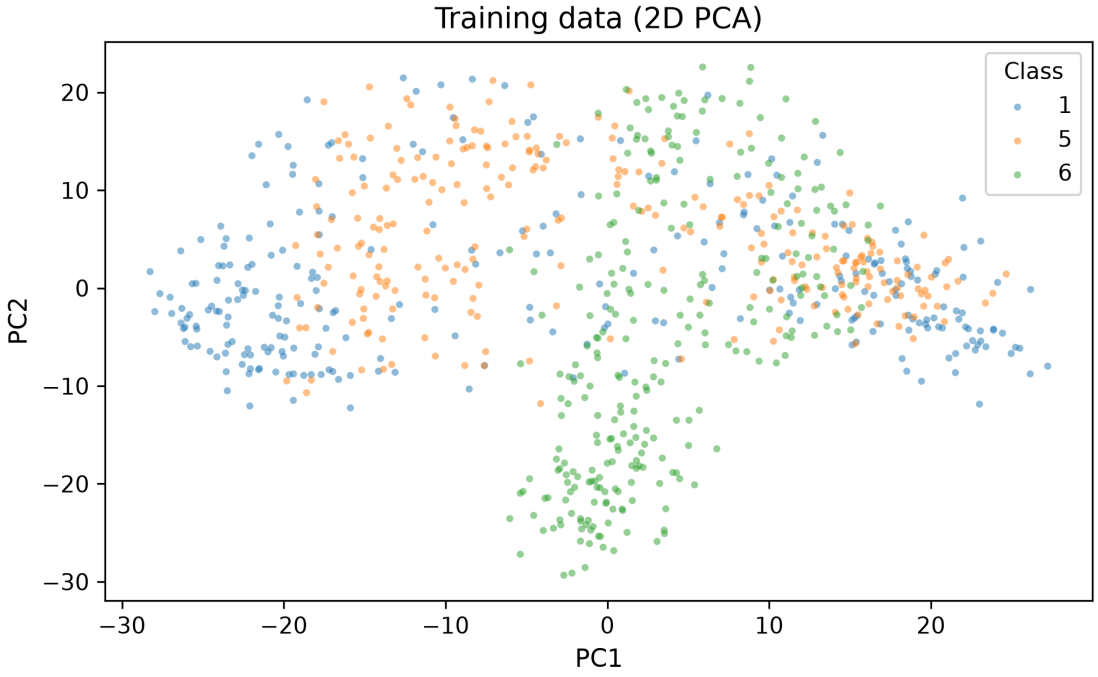
SVM	2	3	9
2	1029	0	3
3	3	1004	3
9	0	7	1002

Observations

All of the four classifiers improved on the test after PCA to 50 dimensions, we believe that PCA has taken out some noise or unwanted variations along with it when it did the dimensionality reduction. SVM gets the best results on this split.

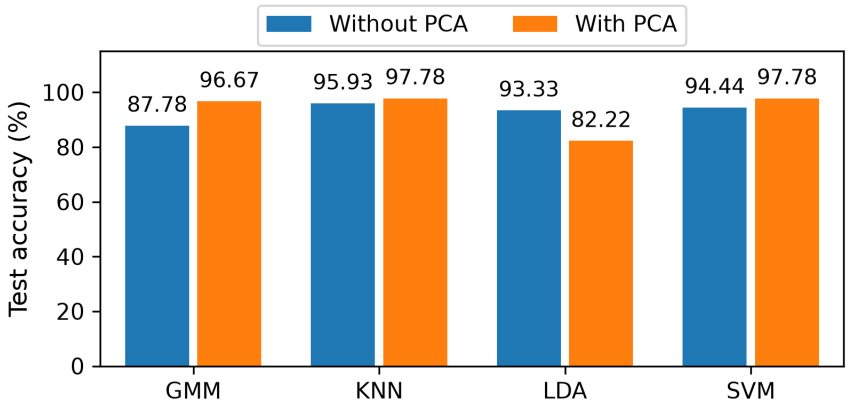
4. TIDIGITS

Labels	Classes	Train	Validation	Test	Features d	PCA k
$\{1, 5, 6\}$	3	810	270	270	4017	39



Accuracy (%)

	Without PCA		With PCA	
Classifier	Validation	Test	Validation	Test
GMM	86.30	87.78	94.44	96.67
KNN	93.33	**95.93**	95.19	**97.78**
LDA	92.22	93.33	83.70	82.22
SVM	94.44	94.44	96.30	**97.78**



Test confusion matrices Rows: actual; columns: predicted.

Without PCA

GMM	1	5	6
1	88	0	0
5	28	64	2
6	3	0	85

KNN	1	5	6
1	88	0	0
5	6	88	0
6	5	0	83

LDA	1	5	6
1	84	4	0
5	9	83	2
6	3	0	85

SVM	1	5	6
1	82	1	5
5	3	86	5
6	0	1	87

With PCA

GMM	1	5	6
1	85	3	0
5	4	89	1
6	0	1	87

KNN	1	5	6
1	86	2	0
5	1	93	0
6	2	1	85

LDA	1	5	6
1	73	15	0
5	17	69	8
6	7	1	80

SVM	1	5	6
1	87	1	0
5	2	91	1
6	1	1	86

Observations

PCA down to 39 dimensions helps GMM, KNN and SVM test accuracy, but LDA drops from 93.33% to 82.22%. We think the answer to this lies in the fact that PCA retains the high-variance directions. But, that is not the necessarily the best direction for discrimination, May be LDA lost some of it's important directions that is useful to it's discrimination. KNN and SVM tie at 97.78% after PCA.